High-Voltage Dual-Supply Control System

Project Documentation

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1 Project Overview

- Intented working environment is to provide bias voltages to RPC detectors, situated in XRID Lab, Tifr-hyderabad.
- Brief was to build power supply of micro-amps reading precison and dec-voltage writing precision, goes upto 5kV inividually and to be controlled remotely as well as manually.
- USB debugging is enabled .Ethernet debugging is under developed.

2 System and Hardware

Hardware blocks

- We used Wiznet evb pico ,has 133 MHz clock freq. RP2040 processor and W5100S Ethernet module. Higher clock freq. helped us with lower latency in comm. and higher display refresh rate. Earlier plan was to go with esp32 40Mhz board, mainly due to its enabled wifi, ble feature. Later shifted to wired comm.
- DAC used is DFRobot Gravity GP8413 here. 15 bit dac with two channels, capable of 0-5V aswell 0-10V output(we used 0-5V).We used only One.Earlier were using 12 bit MCP7425 board, <there was thermal error of 30mV.So we discarded>
- ADC used here is ADS1115 16-Bit,4 Channel(thinner board) , takes analog signals from iseg module's V_vset, V_vmon, I_vmon.We used two of them.It shows 3mV thermal deviation.
- iseg_positive and iseg_negative, we have two each capable of producing 6kV at 6W,max Current of 470uA. Both the module's pins works exactly same except the output polarity is different.
- 2.4inch TFT SPI 240x320 TFT display with no touch sensor(one can try sdcard feature) is used here.SPI protocol helped in higher refresh rate.Unlike LCD display, background is pitch black which enhances display's aesthetic. It shows voltage value of Vset, Vdac, Vmon, Imon for both polarity and debug status.

Signals connection

- DAC(0x59) and ADCs(0x49, 0x48) are connected through I2C. TFT uses SPI protocol. Ethernet module too used SPI protocol. peripherals are connected using jumper wires bunched togeather with tape.
- linear regulator,LM7805 generates 5V using 12V supply which is coming from External DC-DC isolated powersupply situated in the box.
- This 5V provide vcc to all the devices(excluding leds and potentiometers) including microcontroller.
- All devices have same ground including Chasis ground.

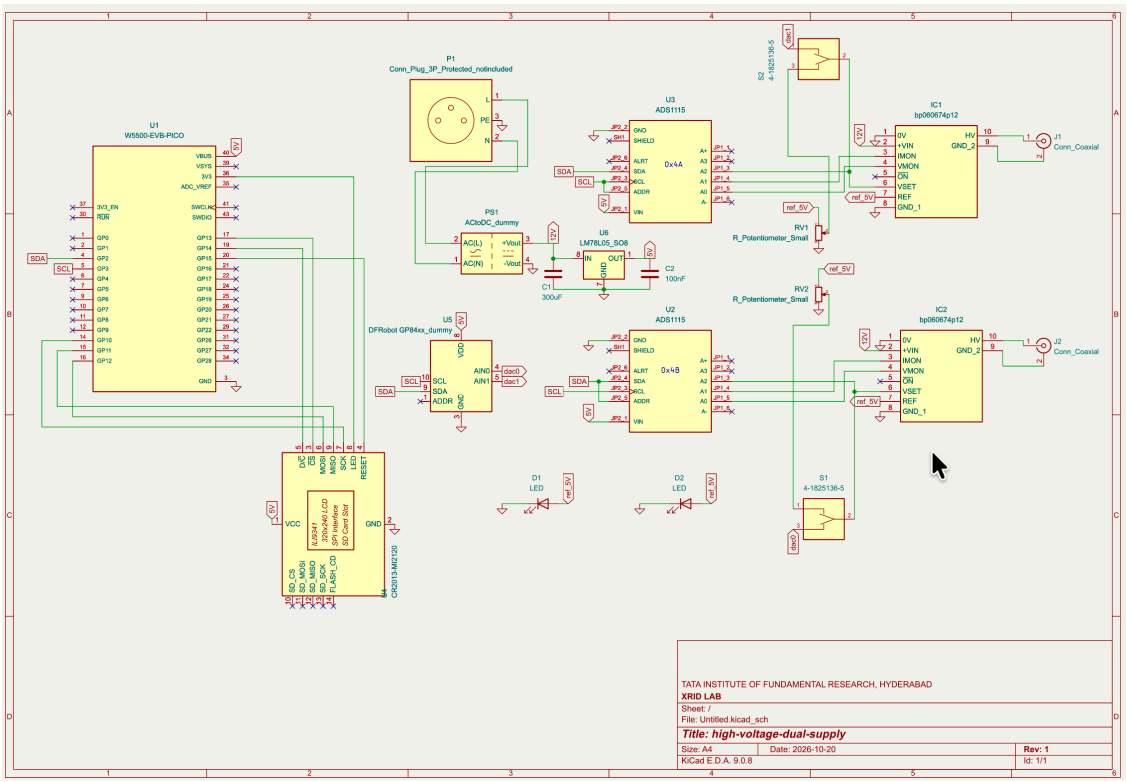


Figure 1: System Schematic

3 Safety

- Two leds situated at the front panel are conneted to 5V reference of iseg module shows status(ON/OFF).
- Shutdown feture can be implemented using /ON pin.In our prototype , we haven't built anything around it.

4 Setup, Build, and Upload

Run this command to clone repository in your local machine: gitclonehttps://github.com/Shawez17/iseg_6kV_controlSystem.git

- What development tools, board packages, libraries, and versions are required?
- Libraries used are:

```
#include <Arduino.h>
#include <Wire.h>
#include <Adafruit_ADS1X15.h>
#include <DFRobot_GP8XXX.h>
config.h:#include <Arduino.h>
```

```
config.h:#include <Adafruit_ADS1X15.h>
config.h:#include <Adafruit_GFX.h>
config.h:#include <Adafruit_ST7789.h>
config.h:#include <DFRobot_GP8XXX.h>
config.h:#include <SPI.h>
config.h:#include <Wire.h>
control-system.ino:#include <Arduino.h>
control-system.ino:#include "hardware/watchdog.h"
control-system.ino:#include "config.h"
control-system.ino:#include "state.h"
control-system.ino:#include "hardware_i2c.h"
control-system.ino:#include "display_ui.h"
control-system.ino:#include "serial_communication.h"
control-system.ino:#include "ethernet_communication.h"
control-system.ino:#include "error_handling.h"
display_images.h:#include <Arduino.h>
display_ui.cpp:#include "hardware/watchdog.h"
display_ui.cpp:#include "display_ui.h"
display_ui.cpp:#include "display_images.h"
display_ui.cpp:#include "error_handling.h"
display_ui.cpp:#include "ethernet_communication.h"
display_ui.cpp:#include "tifrh_logo.h"
display_ui.h:#include <Arduino.h>
display_ui.h:#include <Adafruit_GFX.h>
display_ui.h:#include <Adafruit_ST7789.h>
display_ui.h:#include "state.h"
error_handling.cpp:#include "error_handling.h"
error_handling.cpp:#include "config.h"
error_handling.cpp:#include "display_ui.h"
error_handling.h:#include <Arduino.h>
error_handling.h:#include <Adafruit_ST7789.h>
ethernet_communication.cpp:#include "hardware/watchdog.h"
ethernet_communication.cpp:#include "ethernet_communication.h"
ethernet_communication.cpp:#include <SPI.h>
ethernet_communication.cpp:#include <Ethernet.h>
ethernet_communication.cpp:#include "config.h"
ethernet_communication.cpp:#include "error_handling.h"
ethernet_communication.cpp:#include "serial_communication.h"
ethernet_communication.h:#include <Arduino.h>
ethernet_communication.h:#include "state.h"
hardware_i2c.cpp:#include "hardware/watchdog.h"
hardware_i2c.cpp:#include "hardware_i2c.h"
hardware_i2c.h:#include "config.h"
hardware_i2c.h:#include "state.h"
hardware_i2c.h:#include <Arduino.h>
hardware_i2c.h:#include <Wire.h>
hardware_i2c.h:#include <Adafruit_ADS1X15.h>
hardware_i2c.h:#include <DFRobot_GP8XXX.h>
grep: images: Is a directory
serial_communication.cpp:#include "serial_communication.h"
serial_communication.cpp:#include <Ethernet.h>
serial_communication.cpp:#include "config.h"
serial_communication.cpp:#include "error_handling.h"
serial_communication.h:#include <Arduino.h>
```

```
serial_communication.h:#include "state.h"
state.h:#include <Arduino.h>
state.h:#include "config.h"
tifrh_logo.h:#include <Arduino.h>
```

- Board Url: https://github.com/earlephilhower/arduino-pico/releases/download/global/package_rp2040_index.json
- Command to compile ,flash firmware and monitor using cli:

```
arduino-cli compile --fqbn rp2040:rp2040:wiznet_5100s_evb_pico --output-dir
    build .
arduino-cli upload -p </dev/cu.xxx> --fqbn rp2040:rp2040:
   wiznet_5100s_evb_pico
arduino-cli monitor -p </dev/cu.xxx> -c baudrate=9600
```

- We Used, fqbn: `rp2040:rp2040:wiznet_5100s_evb_pico` Board firmware
- I2C address not found error, was found by avoid long jumper.

5 Operation

- After turning switch ON, both the led must glow, those leds are indicator of working of iseg modules
- TFT Display shows high voltage output around (3-4V), given that potentiometer is at zero position
- Vmon and Vset show no more than 1mV difference
- Two toggle switch exists, one having 6 pins and nother 3 pins. 6pins ts is used for DAC or Potentiometer feature and other one is usb and ethernet debuging option
- One can either watch out readings displayed on the display or pull down the switch ,connect the microcontroller using usb to machine and Monitor it using serial caommands
- If one hears a spark sound constantly , turn off the mains supply and change position of potentiometer to zero(Anti-clock wise).

6 Calibration and Characterization

External Voltage Stability Check

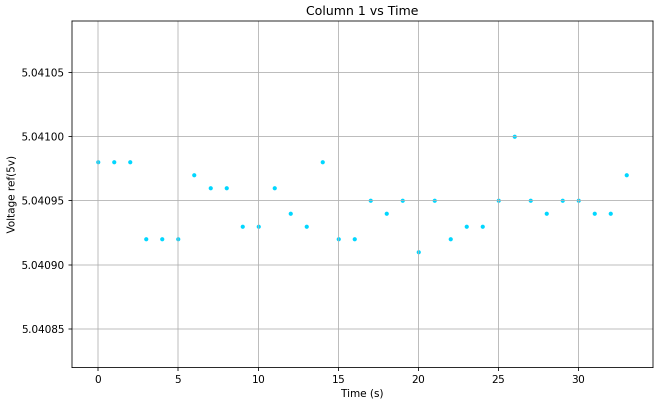


Figure 2: Voltage vs Time(minutes)

MC7525 dev board stability

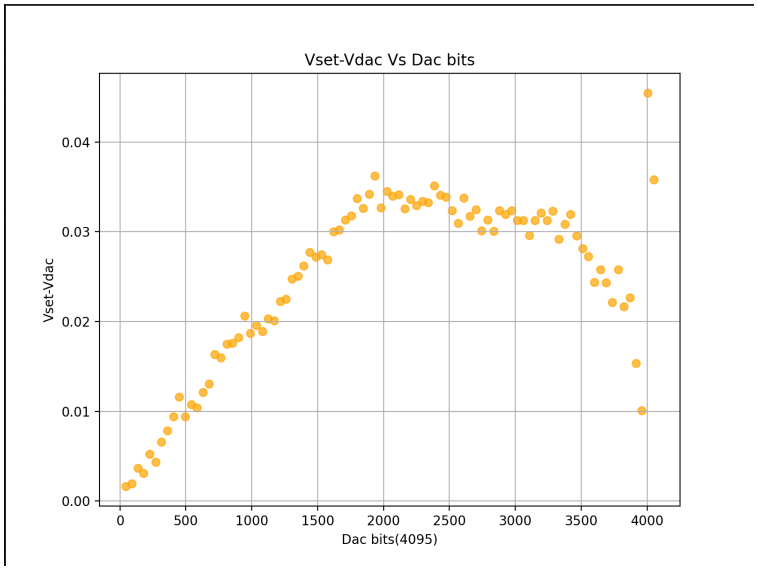


Figure 3: Voltage difference vs dac bits